1 Problem setting

We consider the dynamical field model:

$$u_t = \mathcal{L}(p)u + \mathcal{Q}(u,u) + \mathcal{C}(u,u,u) + \epsilon^n \left(P\mathrm{e}^{\mathrm{i}\omega t} + \overline{P}\mathrm{e}^{-\mathrm{i}\omega t}\right) + \epsilon^m \left(S\mathrm{e}^{\mathrm{i}kx} + \overline{S}\mathrm{e}^{-\mathrm{i}kx}\right), \tag{1}$$

where $u(x,t) \in \mathbb{R}^d$, $x \in \mathbb{T} := \mathbb{R}/(2\pi\mathbb{Z})$, $p \in \mathbb{R}$, and $\mathcal{L}(p) = \mathcal{L}_0 + p\mathcal{L}_1 + O(p^2)$. The last two terms represent the weak forcing. We further assume the model satisfies:

1. $\mathcal{Q}$ and $\mathcal{C}$ are symmetric multilinear maps.
2. $\mathcal{L}_j$ is real, translation invariant, and reflection invariant, $j = 0, 1$.

Since the domain is periodic, every function of x is a linear combination of Fourier modes e^{ihx} where $h \in \mathbb{Z} \setminus \{0\}$. The operators $\mathcal{L}_j$, for $j = 0, 1$, are linear, hence for a spatial Fourier mode, define

$$\mathcal{L}_j\left(v\mathrm{e}^{\mathrm{i}hx}\right) = M_j(h)v\mathrm{e}^{\mathrm{i}hx}, \qquad M_j(h) \in \mathbb{C}^{d\times d}, \qquad h \in \mathbb{Z}\setminus\{0\}. \tag{2}$$

Due to reality and reflection symmetry, respectively,

$$M_j(-h) = \overline{M_j(h)}, \qquad M_j(-h) = M_j(h). \tag{3}$$

Thus $M_j(h)$ is real in this setting and the matrices at h and $-h$ are identical.

1.1 Bifurcation assumption

Now, further assume the Hopf bifurcation scenario: at $p = 0$, all eigenvalues of $M_0(h)$ are nonzero, except for one conjugate pair of eigenvalues for an unique spatial frequency $k_0 \in \mathbb{N}$, so that

$$M_0(\pm k_0)\Phi = \mathrm{i}\omega_0\Phi, \qquad M_0(\pm k_0)^*\Psi = -\mathrm{i}\omega_0\Psi, \qquad \Psi^*\Phi = 1. \tag{4}$$

Here, $\Phi, \Psi \in \mathbb{C}^d$ are corresponding right and adjoint eigenvectors. Hence the only critical modes are the Hopf modes $(\Omega, h) = (\pm\omega_0, k_0), (\pm\omega_0, -k_0)$, where $\omega_0 > 0$ and $k_0 \in \mathbb{N}$. The conjugate equations give the eigenvectors $\overline{\Phi}$ and $\overline{\Psi}$ at eigenvalues $-\mathrm{i}\omega_0$ and $+\mathrm{i}\omega_0$ of M_0 and M_0^*, respectively. Because of (3), the same Φ, Ψ can be used at $\pm k_0$.

Let $\nu(h,p)$ be the eigenvalue branch through $\nu(\pm k_0, 0) = \mathrm{i}\omega_0$. If $M(\pm k_0, p)\Phi(p) = \nu(\pm k_0, p)\Phi(p)$, differentiation at $p = 0$ gives

$$M_1(\pm k_0)\Phi + M_0(\pm k_0)\Phi_p = \nu_p\Phi + \mathrm{i}\omega_0\Phi_p. \tag{5}$$

Since $\Psi^* M_0(\pm k_0) = \mathrm{i}\omega_0\Psi^*$, multiplication by Ψ^* cancels both terms containing Φ_p. Therefore

$$\partial_p\nu(\pm k_0, 0) = \Psi^* M_1(\pm k_0)\Phi =: \alpha, \qquad \operatorname{Re}\alpha > 0. \tag{6}$$

The last inequality is to set p as the bifurcation parameter. $\alpha \neq 0$ will only be used.

Furthermore, for $(\Omega, h) \notin \{(\pm\omega_0, \pm k_0)\}$, define the modal resolvent:

$$R(\Omega, h) := \left(\mathrm{i}\Omega I - M_0(h)\right)^{-1}. \tag{7}$$

Because the system is real, $M_0(-h) = \overline{M_0(h)}$, we have $R(-\Omega, -h) = \overline{R(\Omega, h)}$. In particular,

$$R(-\omega, 0) = \overline{R(\omega, 0)}, \qquad R(0, -k) = \overline{R(0, k)}. \tag{8}$$

With the reflection symmetry assumed here, $M_0(-k) = M_0(k)$ as well, so $R(0,-k) = R(0,k) = \overline{R(0,k)}$. Since $(\Omega, h) = (\pm\omega_0, k_0), (\pm\omega_0, -k_0)$ are only modes, we assume no quadratic resonance:

$$2\mathrm{i}\omega_0 I - M_0(0), \quad 2\mathrm{i}\omega_0 I - M_0(\pm 2k_0), -M_0(0), \quad -M_0(\pm 2k_0) \quad \text{are invertible.} \tag{9}$$

We additionally assume that $R(\pm\omega, 0)$ and $R(0, \pm k)$ exist when the corresponding forcing is present.

1.2 Scales

We introduce fast and slow times

$$s = t, \qquad \tau = \epsilon^2 t, \qquad \partial_t = \partial_s + \epsilon^2 \partial_\tau, \tag{10}$$

and initially write

$$p = \epsilon p_1 + \epsilon^2 p_2 + O(\epsilon^3), \qquad u = \epsilon u_1 + \epsilon^2 u_2 + \epsilon^3 u_3 + O(\epsilon^4). \tag{11}$$

The terms that determine the first three equations are

$$\begin{aligned}
\partial_t u = &\quad \epsilon\, \partial_s u_1 &&+\epsilon^2\, \partial_s u_2 &&+\epsilon^3 \big(\partial_s u_3 + \partial_\tau u_1\big) + O(\epsilon^4),\\
\mathcal{L}_0 u = &\quad \epsilon\, \mathcal{L}_0 u_1 &&+\epsilon^2\, \mathcal{L}_0 u_2 &&+\epsilon^3\, \mathcal{L}_0 u_3 + O(\epsilon^4),\\
p\mathcal{L}_1 u = &&&\epsilon^2 p_1 \mathcal{L}_1 u_1 &&+\epsilon^3 \big(p_2 \mathcal{L}_1 u_1 + p_1 \mathcal{L}_1 u_2\big) + O(\epsilon^4),\\
\mathcal{Q}(u,u) = &&&\epsilon^2 \mathcal{Q}(u_1,u_1) &&+2\epsilon^3 \mathcal{Q}(u_1,u_2) + O(\epsilon^4),\\
\mathcal{C}(u,u,u) = &&&&&\epsilon^3 \mathcal{C}(u_1,u_1,u_1) + O(\epsilon^4).
\end{aligned} \tag{12}$$

2 The solution space of $\mathcal{D} = \partial_s - \mathcal{L}_0$

First, let's define the linear operator and wave forms:

$$\mathcal{D} := \partial_s - \mathcal{L}_0, \qquad E_\pm := \mathrm{e}^{\mathrm{i}(\omega_0 s \pm k_0 x)}. \tag{13}$$

For any pure wave,

$$\mathcal{D}\left(v\mathrm{e}^{\mathrm{i}(\Omega s + hx)}\right) = (\mathrm{i}\Omega I - M_0(h))\, v\mathrm{e}^{\mathrm{i}(\Omega s + hx)}. \tag{14}$$

From the assumption, the multiplier in (14) is only singular at $(\Omega, h) = (\omega_0, \pm k_0)$ with unique nullvector Φ, or its conjugate which is singular at $(-\omega_0, \mp k_0)$ with nullvector $\overline{\Phi}$. Hence,

$$\ker \mathcal{D} = \mathrm{span}_{\mathbb{R}} \left\{ \Phi E_+, \Phi E_-, \overline{\Phi}\, \overline{E}_+, \overline{\Phi}\, \overline{E}_- \right\}. \tag{15}$$

2.1 Inner product

On functions periodic in x and in s with $T_0 = 2\pi/\omega_0$, use

$$\langle v, w \rangle := \frac{1}{2\pi T_0} \int_0^{T_0} \int_0^{2\pi} v(s,x)^* w(s,x)\, dx\, ds. \tag{16}$$

Periodic integration by parts gives

$$\mathcal{D}^* = -\partial_s - \mathcal{L}_0^*, \qquad \mathcal{D}^*\left(Z\mathrm{e}^{\mathrm{i}(\Omega s + hx)}\right) = (-\mathrm{i}\Omega I - M_0(h)^*)\, Z\mathrm{e}^{\mathrm{i}(\Omega s + hx)}. \tag{17}$$

So (4) gives the kernel space of $\mathcal{D}^*$:

$$\ker \mathcal{D}^* = \mathrm{span}_{\mathbb{R}} \left\{ \Psi E_+, \Psi E_-, \overline{\Psi}\, \overline{E}_+, \overline{\Psi}\, \overline{E}_- \right\}. \tag{18}$$

For waves that are Fourier basis,

$$\left\langle a\mathrm{e}^{\mathrm{i}(\Omega s + hx)}, b\mathrm{e}^{\mathrm{i}(\Omega' s + h' x)} \right\rangle = a^* b\, \delta_{\Omega,\Omega'} \delta_{h,h'}. \tag{19}$$

Thus different wave types are orthogonal. Having the same frequency and wavenumber pair is necessary for a nonzero inner product.

2.2 Fredholm alternative

For the notational convenience, let

$$U_\pm = \Phi E_\pm, \qquad V_\pm = \Psi E_\pm. \tag{20}$$

The complete pairings needed below are

	$\mathcal{L}_1 U_+$	$\mathcal{L}_1 U_-$	$\mathcal{L}_1 \overline{U}_+$	$\mathcal{L}_1 \overline{U}_-$
V_+	α	0	0	0
V_-	0	α	0	0
$\overline{V}_+$	0	0	$\overline{\alpha}$	0
$\overline{V}_-$	0	0	0	$\overline{\alpha}$

. (21)

Using the Fredholm alternative,

$$\operatorname{Ran}\mathcal{D} = (\ker \mathcal{D}^*)^\perp, \qquad \mathcal{D}w = f \iff \langle V, f\rangle = 0 \quad \text{for every } V \in \ker \mathcal{D}^*. \tag{22}$$

At every order, the goal is to put the right-hand side in $\operatorname{Ran}\mathcal{D}$.

2.3 Forcing scales

To prevent the stimulus itself from supplying the leading pattern, take $n, m > 1$. If $n = 1$ or $m = 1$, the corresponding stimulus appears in $\mathcal{D}u_1$ and u_1 is no longer chosen solely from $\ker\mathcal{D}$. If an exponent equals 2, its response is part of u_2 and interacts with u_1 through $2\mathcal{Q}(u_1, u_2)$ at order three. An exponent greater than 2 cannot produce such a cubic-order feedback. Thus, we only consider

$$F_2 = \mathbf{1}_{\{n=2\}}\left(P\mathrm{e}^{\mathrm{i}\omega s} + \overline{P}\mathrm{e}^{-\mathrm{i}\omega s}\right) + \mathbf{1}_{\{m=2\}}\left(S\mathrm{e}^{\mathrm{i}kx} + \overline{S}\mathrm{e}^{-\mathrm{i}kx}\right). \tag{23}$$

The three equations at each scales are

$$O(\epsilon): \quad \mathcal{D}u_1 = 0, \tag{24a}$$

$$O(\epsilon^2): \quad \mathcal{D}u_2 = p_1\mathcal{L}_1 u_1 + \mathcal{Q}(u_1, u_1) + F_2, \tag{24b}$$

$$O(\epsilon^3): \quad \mathcal{D}u_3 = -\partial_\tau u_1 + p_2\mathcal{L}_1 u_1 + 2\mathcal{Q}(u_1, u_2) + \mathcal{C}(u_1, u_1, u_1). \tag{24c}$$

3 $O(\epsilon)$ solution

From the equation

$$\mathcal{D}u_1 = 0,$$

the real leading solution is

$$\boxed{u_1 = A(\tau)\Phi E_+ + B(\tau)\Phi E_- + \text{c.c.}.} \tag{25}$$

4 $O(\epsilon^2)$ solution

The system at $O(\epsilon^2)$ level is:

$$\mathcal{D}u_2 = p_1\mathcal{L}_1 u_1 + \mathcal{Q}(u_1, u_1) + F_2.$$

There are wave types of $\{-2\omega_0, 0, 2\omega_0\} \times \{-2k_0, 0, 2k_0\}$ in the term $\mathcal{Q}(u_1, u_1)$ and $\{(\pm\omega, 0), (0, \pm k)\}$ in the term F_2. Hence, $p_1\mathcal{L}_1 u_1$ need to satisfy the Fredholm alternative.

4.1 Fredholm alternative check

The right-hand side of (24b) must be orthogonal to $\ker \mathcal{D}^*$. From the wave types of $\mathcal{Q}(u_1, u_1)$,

$$\langle V_\pm, \mathcal{Q}(u_1, u_1)\rangle = 0. \tag{26}$$

Similarly, for the forcing, $\langle V_\pm, F_2\rangle = 0$. Hence, only $p_1\mathcal{L}_1 u_1$ remains. By (21),

$$\langle V_+, p_1\mathcal{L}_1 u_1\rangle = p_1 A\alpha, \qquad \langle V_-, p_1\mathcal{L}_1 u_1\rangle = p_1 B\alpha. \tag{27}$$

Thus the two independent complex conditions are

$$0 = p_1 A\alpha, \qquad 0 = p_1 B\alpha. \tag{28}$$

For $(A, B) \neq (0, 0)$ and $\operatorname{Re}\alpha > 0$,

$$\boxed{p_1 = 0, \qquad p = \epsilon^2\mu + O(\epsilon^3).} \tag{29}$$

Hence, the system at $O(\epsilon^2)$ level is:

$$\mathcal{D}u_2 = \mathcal{Q}(u_1, u_1) + F_2.$$

4.2 Term $u_2^Q := \mathcal{D}^{-1}\mathcal{Q}(u_1, u_1)$

Set the u_1 function as:

$$u_1^+ := \Phi(AE_+ + BE_-), \qquad u_1 = u_1^+ + \overline{u_1^+}. \tag{30}$$

Symmetry of $\mathcal{Q}$ gives

$$\mathcal{Q}(u_1, u_1) = \mathcal{Q}(u_1^+, u_1^+) + 2\mathcal{Q}(u_1^+, \overline{u_1^+}) + \mathcal{Q}(\overline{u_1^+}, \overline{u_1^+}). \tag{31}$$

Therefore, with $q_{20} := \mathcal{Q}(\Phi, \Phi)$ and $q_{11} := \mathcal{Q}(\Phi, \overline{\Phi})$,

$$\begin{aligned}\mathcal{Q}(u_1, u_1) = {} & q_{20}\left(A^2E_+^2 + B^2E_-^2 + 2ABE_+E_-\right) + \text{c.c.} \\ & + 2q_{11}\left(|A|^2 + |B|^2 + A\overline{B}E_+\overline{E}_- + \overline{A}B\overline{E}_+E_-\right).\end{aligned} \tag{32}$$

The generated wave types are

$$(\pm 2\omega_0, \pm 2k_0), \quad (\pm 2\omega_0, 0), \quad (0, \pm 2k_0), \quad (0, 0).$$

The condition (9) is the additional statement that $\mathcal{D}$ is invertible on each of them. Recall:

$$\mathcal{D}^{-1}\left(fe^{\mathrm{i}(\Omega s + hx)}\right) = R(\Omega, h)fe^{\mathrm{i}(\Omega s + hx)}.$$

Set $R\mathcal{Q}$ for each case as:

$$\begin{aligned} &h_{20}^+ = R(2\omega_0, 2k_0)q_{20}, && h_{20}^- = R(2\omega_0, -2k_0)q_{20}, && h_{20}^0 = 2R(2\omega_0, 0)q_{20}, \\ &h_{11}^0 = 2R(0, 0)q_{11}, && h_{11}^+ = 2R(0, 2k_0)q_{11}, && h_{11}^- = 2R(0, -2k_0)q_{11}. \end{aligned}$$

Reflection symmetry gives $h_{20}^+ = h_{20}^-$ and $h_{11}^+ = h_{11}^-$, but the superscripts are retained to display the wave types. Each terms in the quadratic term and it's effect on u_2 is:

$\mathcal{Q}(u_1, u_1)$	$A^2q_{20}E_+^2$	$B^2q_{20}E_-^2$	$2ABq_{20}E_+E_-$	$2(\lvert A\rvert^2 + \lvert B\rvert^2)q_{11}$	$2A\overline{B}q_{11}E_+\overline{E}_-$	$2\overline{A}Bq_{11}\overline{E}_+E_-$
(Ω, h)	$(2\omega_0, 2k_0)$	$(2\omega_0, -2k_0)$	$(2\omega_0, 0)$	$(0, 0)$	$(0, 2k_0)$	$(0, -2k_0)$
u_2^Q	$A^2h_{20}^+E_+^2$	$B^2h_{20}^-E_-^2$	$ABh_{20}^0E_+E_-$	$(\lvert A\rvert^2 + \lvert B\rvert^2)h_{11}^0$	$A\overline{B}h_{11}^+E_+\overline{E}_-$	$\overline{A}Bh_{11}^-\overline{E}_+E_-$

Hence,

$$\begin{aligned}u_2^Q := \mathcal{D}^{-1}\mathcal{Q}(u_1, u_1) = {} & A^2h_{20}^+E_+^2 + B^2h_{20}^-E_-^2 + ABh_{20}^0E_+E_- + \text{c.c.} \\ & + (|A|^2 + |B|^2)h_{11}^0 + A\overline{B}h_{11}^+E_+\overline{E}_- + \overline{A}Bh_{11}^-\overline{E}_+E_-.\end{aligned} \tag{33}$$

4.3 Term $u_2^F := \mathcal{D}^{-1}F_2$

For the the forcing frequency and wavenumber, define:

$$p_\omega := R(\omega, 0)P, \qquad s_k := R(0, k)S. \tag{34}$$

The symmetry (8) gives

$$R(-\omega, 0)\overline{P} = \overline{p_\omega}, \qquad R(0, -k)\overline{S} = \overline{s_k}. \tag{35}$$

By taking the inverse operator of $\mathcal{D}$, we get

$$u_2^F := \mathcal{D}^{-1}F_2 = \mathbf{1}_{\{n=2\}}\left(p_\omega \mathrm{e}^{\mathrm{i}\omega s} + \overline{p_\omega}\mathrm{e}^{-\mathrm{i}\omega s}\right) + \mathbf{1}_{\{m=2\}}\left(s_k \mathrm{e}^{\mathrm{i}kx} + \overline{s_k}\mathrm{e}^{-\mathrm{i}kx}\right). \tag{36}$$

The conjugate terms make u_2^F real. The $O(\epsilon^2)$ solution therefore is:

$$\boxed{u_2 = u_2^Q + u_2^F.} \tag{37}$$

if there are no added components from $\ker \mathcal{D}$.

5 $O(\epsilon^3)$ solution and solvability

With $p_2 = \mu$, write the right-hand side of (24c) in two groups:

$$\mathcal{D}u_3 = \underbrace{-\partial_\tau u_1 + \mu\mathcal{L}_1 u_1 + 2\mathcal{Q}(u_1, u_2^Q) + \mathcal{C}(u_1, u_1, u_1)}_{=:F_0^{(3)}} + \underbrace{2\mathcal{Q}(u_1, u_2^F)}_{=:F_{Force}^{(3)}}. \tag{38}$$

In this section, we focus on the coefficients of the critical wave E_+, to analyze the solvability with Fredholm alternative. Note that for other critical waves, coefficients will be similar.

5.1 Unforced terms $F_0^{(3)}$

For the critical wave E_+, the linear term gives coefficients

$$[-\partial_\tau u_1 + \mu\mathcal{L}_1 u_1]_{E_+} = -A_\tau \Phi + \mu A M_1(k_0)\Phi. \tag{39}$$

For the cubic term, a triple of critical phases can sum to E_+ only as $E_+E_+\overline{E}_+$ or $E_+E_-\overline{E}_-$:

$$[\mathcal{C}(u_1, u_1, u_1)]_{E_+} = 3A|A|^2\mathcal{C}(\Phi, \Phi, \overline{\Phi}) + 6A|B|^2\mathcal{C}(\Phi, \Phi, \overline{\Phi}), \tag{40}$$

where 3 and 6 are the permutation counts. For the quadratic term, E_+ emerges at:

u_1 terms	$A\Phi E_+$	$\overline{A}\,\overline{\Phi}\,\overline{E}_+$	$\overline{B}\,\overline{\Phi}\,\overline{E}_-$	$B\Phi E_-$
u_2^Q terms	$(\lvert A\rvert^2 + \lvert B\rvert^2)h_{11}^0$	$A^2 h_{20}^+ E_+^2$	$AB h_{20}^0 E_+ E_-$	$A\overline{B}h_{11}^+ E_+\overline{E}_-$
$2\mathcal{Q}$ terms	$2A(\lvert A\rvert^2 + \lvert B\rvert^2)\mathcal{Q}(\Phi, h_{11}^0)$	$2A\lvert A\rvert^2\mathcal{Q}(\overline{\Phi}, h_{20}^+)$	$2A\lvert B\rvert^2\mathcal{Q}(\overline{\Phi}, h_{20}^0)$	$2A\lvert B\rvert^2\mathcal{Q}(\Phi, h_{11}^+)$

No other pair of wave types sums to (ω_0, k_0). Summing the last column gives

$$[2\mathcal{Q}(u_1, u_2^Q)]_{E_+} = 2A(|A|^2 + |B|^2)\mathcal{Q}(\Phi, h_{11}^0) + 2A|A|^2\mathcal{Q}(\overline{\Phi}, h_{20}^+) \tag{41}$$

$$+ 2A|B|^2\mathcal{Q}(\overline{\Phi}, h_{20}^0) + 2A|B|^2\mathcal{Q}(\Phi, h_{11}^+). \tag{42}$$

For the simplicity, define two coefficient terms for $A|A|^2$ and $A|B|^2$:

$$G_{\mathrm{s}} = 2\mathcal{Q}(\Phi, h_{11}^0) + 2\mathcal{Q}(\overline{\Phi}, h_{20}^+) + 3\mathcal{C}(\Phi, \Phi, \overline{\Phi}), \tag{43a}$$

$$G_{\mathrm{c}} = 2\mathcal{Q}(\Phi, h_{11}^0) + 2\mathcal{Q}(\overline{\Phi}, h_{20}^0) + 2\mathcal{Q}(\Phi, h_{11}^+) + 6\mathcal{C}(\Phi, \Phi, \overline{\Phi}). \tag{43b}$$

With the same steps on E_-, we get similar coefficients with A and B are exchanged. Therefore

$$[F_0^{(3)}]_{E_+} = -A_\tau\Phi + \mu A M_1(k_0)\Phi + A|A|^2 G_{\mathrm{s}} + A|B|^2 G_{\mathrm{c}}, \tag{44}$$

$$[F_0^{(3)}]_{E_-} = -B_\tau\Phi + \mu B M_1(k_0)\Phi + B|B|^2 G_{\mathrm{s}} + B|A|^2 G_{\mathrm{c}}. \tag{45}$$

These and their conjugates are the only unforced terms outside $\operatorname{Ran}\mathcal{D}$.

5.2 Forcing-interaction terms $F_{Force}^{(3)}$

Recall that

$$u_1 = A\Phi E_+ + B\Phi E_- + \bar{A}\bar{\Phi}\bar{E_+} + \bar{B}\bar{\Phi}\bar{E_-},$$

and

$$u_2^F = u_{2,t}^F + u_{2,x}^F, \quad u_{2,t}^F \coloneqq \mathbf{1}_{\{n=2\}}\left(p_\omega \mathrm{e}^{\mathrm{i}\omega s} + \overline{p_\omega}\mathrm{e}^{-\mathrm{i}\omega s}\right), \quad u_{2,x}^F \coloneqq \mathbf{1}_{\{m=2\}}\left(s_k \mathrm{e}^{\mathrm{i}kx} + \overline{s_k}\mathrm{e}^{-\mathrm{i}kx}\right).$$

Hence, the wave types falls into the form of one of the followings:

$$\exp\left[\pm\mathrm{i}((\omega_0 \pm \omega)s \pm k_0 x)\right], \qquad \exp\left[\pm\mathrm{i}(\omega_0 s \pm (k_0 \pm k)x)\right],$$

where each $\pm$ are not correlated. To have nonzero critical wave terms, the only possible case is when $\omega = 0, \pm 2\omega_0$ and $k = 0, \pm 2k_0$. With $\omega, k \geq 0$, we have coefficients for the temporal forcing effects:

$$[2\mathcal{Q}(u_1, u_{2,t}^F)]_{E_+} = \mathbf{1}_{\{n=2\}}\left[\mathbf{1}_{\{\omega=0\}} 2A\mathcal{Q}(\Phi, p_0 + \overline{p_0}) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{B}\mathcal{Q}(\overline{\Phi}, p_{2\omega_0})\right], \tag{46}$$

$$[2\mathcal{Q}(u_1, u_{2,t}^F)]_{E_-} = \mathbf{1}_{\{n=2\}}\left[\mathbf{1}_{\{\omega=0\}} 2B\mathcal{Q}(\Phi, p_0 + \overline{p_0}) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{A}\mathcal{Q}(\overline{\Phi}, p_{2\omega_0})\right]. \tag{47}$$

In the same way, we have coefficients for the spatial forcing effects:

$$[2\mathcal{Q}(u_1, u_{2,x}^F)]_{E_+} = \mathbf{1}_{\{m=2\}}\left[\mathbf{1}_{\{k=0\}} 2A\mathcal{Q}(\Phi, s_0 + \overline{s_0}) + \mathbf{1}_{\{k=2k_0\}} 2B\mathcal{Q}(\Phi, s_{2k_0})\right], \tag{48}$$

$$[2\mathcal{Q}(u_1, u_{2,x}^F)]_{E_-} = \mathbf{1}_{\{m=2\}}\left[\mathbf{1}_{\{k=0\}} 2B\mathcal{Q}(\Phi, s_0 + \overline{s_0}) + \mathbf{1}_{\{k=2k_0\}} 2A\mathcal{Q}(\Phi, \overline{s_{2k_0}})\right]. \tag{49}$$

5.3 Fredholm conditions

Using $\Psi^*\Phi = 1$, the projection to the kernel space of $\mathcal{D}^*$ is:

$$\left\langle V_+, F^{(3)}\right\rangle = \Psi^*[F^{(3)}]_{E_+}, \qquad \left\langle V_-, F^{(3)}\right\rangle = \Psi^*[F^{(3)}]_{E_-}. \tag{50}$$

Define values

$$a \coloneqq \mu\Psi^* M_1(k_0)\Phi = \mu\alpha, \qquad b \coloneqq \Psi^* G_{\mathrm{s}}, \qquad c \coloneqq \Psi^* G_{\mathrm{c}}, \tag{51}$$

and denote forcing projections as:

$$\begin{aligned}
\mathcal{T}_A &= \mathbf{1}_{\{n=2\}}\left[\mathbf{1}_{\{\omega=0\}} 2A\Psi^*\mathcal{Q}(\Phi, p_0 + \overline{p_0}) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{B}\Psi^*\mathcal{Q}(\overline{\Phi}, p_{2\omega_0})\right], \\
\mathcal{T}_B &= \mathbf{1}_{\{n=2\}}\left[\mathbf{1}_{\{\omega=0\}} 2B\Psi^*\mathcal{Q}(\Phi, p_0 + \overline{p_0}) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{A}\Psi^*\mathcal{Q}(\overline{\Phi}, p_{2\omega_0})\right], \\
\mathcal{X}_A &= \mathbf{1}_{\{m=2\}}\left[\mathbf{1}_{\{k=0\}} 2A\Psi^*\mathcal{Q}(\Phi, s_0 + \overline{s_0}) + \mathbf{1}_{\{k=2k_0\}} 2B\Psi^*\mathcal{Q}(\Phi, s_{2k_0})\right], \\
\mathcal{X}_B &= \mathbf{1}_{\{m=2\}}\left[\mathbf{1}_{\{k=0\}} 2B\Psi^*\mathcal{Q}(\Phi, s_0 + \overline{s_0}) + \mathbf{1}_{\{k=2k_0\}} 2A\Psi^*\mathcal{Q}(\Phi, \overline{s_{2k_0}})\right].
\end{aligned}$$

Then, the Fredholm alternative now gives

$$0 = -A_\tau + A(a + b|A|^2 + c|B|^2) + \mathcal{T}_A + \mathcal{X}_A, \tag{52a}$$

$$0 = -B_\tau + B(a + b|B|^2 + c|A|^2) + \mathcal{T}_B + \mathcal{X}_B. \tag{52b}$$

6 Amplitude equations and patterns

Considering only non-homogeneous forcing at $2\omega_0$ and $2k_0$, define the complex forcing coefficients

$$\eta_t := 2\Psi^* \mathcal{Q}\big(\overline{\Phi}, p_{2\omega_0}\big)\,, \qquad \eta_+ := 2\Psi^* \mathcal{Q}(\Phi, s_{2k_0})\,, \qquad \eta_- := 2\Psi^* \mathcal{Q}(\Phi, \overline{s_{2k_0}})\,.$$

Thus $\eta_t = 0$ when $P = 0$, and $\eta_x = 0$ when $S = 0$. The amplitude equations are

$$A_\tau = A\left(a + b|A|^2 + c|B|^2\right) + \eta_t \overline{B} + \eta_+ B, \tag{53a}$$
$$B_\tau = B\left(a + b|B|^2 + c|A|^2\right) + \eta_t \overline{A} + \eta_- A. \tag{53b}$$

Here $a, b, c, \eta_t, \eta_x \in \mathbb{C}$. An equilibrium satisfies

$$0 = A\left(a + b|A|^2 + c|B|^2\right) + \eta_t \overline{B} + \eta_+ B, \tag{54}$$
$$0 = B\left(a + b|B|^2 + c|A|^2\right) + \eta_t \overline{A} + \eta_- A. \tag{55}$$

6.1 Existence of a nonzero pattern below onset

Take $n = m = 2$, $\omega = 2\omega_0$, $k = 2k_0$, and

$$p = \epsilon^2 \mu, \qquad \mu < 0, \qquad a = \mu\alpha. \tag{56}$$

All scaled coefficients are fixed as $\epsilon \to 0$. Assume $S \in \mathbb{R}^d$, so that s_{2k_0} is real and $\eta_+ = \eta_- =: \eta_x$. Set

$$a_+ = a + \eta_x, \qquad \gamma = b + c. \tag{57}$$

Since $\operatorname{Re} a = \mu \operatorname{Re} \alpha < 0$, the unforced zero-amplitude solution is linearly stable.

Proposition 1. *If $\gamma \neq 0$ and $|\eta_t| > |a_+|$, then (53) has a nonzero equilibrium $A_* = B_* = z_*$. Its leading field is a standing pattern of order ϵ.*

Proof. The subspace $A = B = z$ is invariant, and its equation is

$$z_\tau = z(a_+ + \gamma|z|^2) + \eta_t \overline{z}. \tag{58}$$

For $z = \sqrt{\rho}\, \mathrm{e}^{\mathrm{i}\phi} \neq 0$, the equilibrium condition gives

$$a_+ + \gamma\rho + \eta_t \mathrm{e}^{-2\mathrm{i}\phi} = 0, \qquad \rho > 0. \tag{59}$$

Taking absolute values from $a_+ + \gamma\rho = -\eta_t \mathrm{e}^{-2\mathrm{i}\phi}$, define

$$f(\rho) := |a_+ + \gamma\rho|^2 - |\eta_t|^2 = |\gamma|^2\rho^2 + 2\operatorname{Re}(a_+\overline{\gamma})\rho + |a_+|^2 - |\eta_t|^2. \tag{60}$$

Its leading coefficient is positive and its constant term is negative. Hence it has exactly one positive root ρ_*, with $f'(\rho_*) > 0$. Choose ϕ_* by

$$\mathrm{e}^{-2\mathrm{i}\phi_*} = -\frac{a_+ + \gamma\rho_*}{\eta_t}, \qquad z_* = \sqrt{\rho_*}\, \mathrm{e}^{\mathrm{i}\phi_*}. \tag{61}$$

The quotient has modulus one, so $A_* = B_* = z_*$ is an equilibrium. The leading field is therefore

$$u(x,t) = 4\epsilon\sqrt{\rho_*}\, \operatorname{Re}\left[\Phi \mathrm{e}^{\mathrm{i}(\omega_0 t + \phi_*)}\right] \cos(k_0 x) + O(\epsilon^2). \tag{62}$$

Since $\rho_* > 0$, $\Phi \neq 0$, and $k_0 \neq 0$, its leading term is nonzero and spatially nonconstant. □

Stability. At $z = 0$, (58) has eigenvalues

$$\lambda_\pm = \operatorname{Re} a_+ \pm \sqrt{|\eta_t|^2 - (\operatorname{Im} a_+)^2}. \tag{63}$$

The condition $|\eta_t| > |a_+|$ gives $\lambda_+ > 0$. Thus the forcing destabilizes zero critical amplitude.

At $(A_*, B_*) = (z_*, z_*)$, write

$$v_\pm = \tfrac{1}{2}\mathrm{e}^{-\mathrm{i}\phi_*}(\delta A \pm \delta B). \tag{64}$$

Using (59), the linearized equations separate as

$$(v_+)_\tau = (a_+ + 2\gamma\rho_*)v_+ - a_+\overline{v}_+, \tag{65}$$

$$(v_-)_\tau = (a - \eta_x + 2b\rho_*)v_- + (a_+ + 2b\rho_*)\overline{v}_-. \tag{66}$$

For $v_\tau = hv + \ell\overline{v}$, the real Jacobian has trace $2\operatorname{Re} h$ and determinant $|h|^2 - |\ell|^2$. Thus

$$\begin{aligned} \operatorname{tr} J_+ &= 2\operatorname{Re}(a_+ + 2\gamma\rho_*), & \det J_+ &= 2\rho_* f'(\rho_*) > 0, \\ \operatorname{tr} J_- &= 2\operatorname{Re}(a - \eta_x + 2b\rho_*), & \det J_- &= -4\operatorname{Re}[(a + 2b\rho_*)\overline{\eta_x}]\,. \end{aligned} \tag{67}$$

Consequently the equilibrium is strictly linearly stable precisely when

$$\operatorname{Re}(a_+ + 2\gamma\rho_*) < 0, \quad \operatorname{Re}(a - \eta_x + 2b\rho_*) < 0, \quad \operatorname{Re}[(a + 2b\rho_*)\overline{\eta_x}] < 0. \tag{68}$$

For pure temporal forcing, $S = 0$ and $\eta_x = 0$. In this case J_- has eigenvalues 0 and $2\operatorname{Re}(a+2b\rho_*)$. The zero eigenvalue is tangent to the translation family

$$(A, B) = (z_*\mathrm{e}^{\mathrm{i}\theta}, z_*\mathrm{e}^{-\mathrm{i}\theta}), \qquad \theta \in \mathbb{R}. \tag{69}$$

Since $\operatorname{Re} a < 0$, the sufficient conditions

$$\boxed{|\eta_t| > |\mu\alpha|, \qquad \operatorname{Re} b < 0, \qquad \operatorname{Re}(b + c) < 0} \tag{70}$$

make the other three eigenvalues have negative real part. The standing wave is therefore locally asymptotically stable modulo spatial translations in the amplitude system.

6.2 Notation summary

Table 1. Shorthand quantities in Sections 1–6.

Symbol	Definition	Location
$M_j(h)$	Fourier matrix: $\mathcal{L}_j(v\mathrm{e}^{\mathrm{i}hx}) = M_j(h)v\mathrm{e}^{\mathrm{i}hx}$.	(2)
Φ, Ψ	Critical right and adjoint eigenvectors; $\Psi^*\Phi = 1$.	(4)
α	$\Psi^* M_1(k_0)\Phi = \partial_p \nu(k_0, 0)$.	(6)
$\mathcal{D}, R(\Omega, h)$	$\mathcal{D} = \partial_s - \mathcal{L}_0$, $R(\Omega, h) = (\mathrm{i}\Omega I - M_0(h))^{-1}$.	(13), (7)
$E_\pm, U_\pm, V_\pm$	$E_\pm = \mathrm{e}^{\mathrm{i}(\omega_0 s \pm k_0 x)}$, $U_\pm = \Phi E_\pm$, $V_\pm = \Psi E_\pm$.	Section 2
A, B	$u_1 = A\Phi E_+ + B\Phi E_- + \text{c.c.}$.	(25)
q_{20}, q_{11}	$q_{20} = \mathcal{Q}(\Phi, \Phi)$, $q_{11} = \mathcal{Q}(\Phi, \overline{\Phi})$.	Section 4.2
$h_{20}^\pm, h_{20}^0$	$h_{20}^\pm = R(2\omega_0, \pm 2k_0)q_{20}$, $h_{20}^0 = 2R(2\omega_0, 0)q_{20}$.	Section 4.2
$h_{11}^\pm, h_{11}^0$	$h_{11}^\pm = 2R(0, \pm 2k_0)q_{11}$, $h_{11}^0 = 2R(0, 0)q_{11}$.	Section 4.2
p_ω, s_k	$p_\omega = R(\omega, 0)P$, $s_k = R(0, k)S$.	(34)
u_2^Q, u_2^F	$u_2^Q = \mathcal{D}^{-1}\mathcal{Q}(u_1, u_1)$, $u_2^F = \mathcal{D}^{-1}F_2$.	(33), (36)
G_s	$2\mathcal{Q}(\Phi, h_{11}^0) + 2\mathcal{Q}(\overline{\Phi}, h_{20}^+) + 3\mathcal{C}(\Phi, \Phi, \overline{\Phi})$.	(43a)
G_c	$2\mathcal{Q}(\Phi, h_{11}^0) + 2\mathcal{Q}(\overline{\Phi}, h_{20}^0) + 2\mathcal{Q}(\Phi, h_{11}^+) + 6\mathcal{C}(\Phi, \Phi, \overline{\Phi})$.	(43b)
a, b, c	$a = \mu\alpha$, $b = \Psi^* G_\mathrm{s}$, $c = \Psi^* G_\mathrm{c}$, with $\mu = p_2$.	(51)
$\mathcal{T}_{A,B}, \mathcal{X}_{A,B}$	Temporal and spatial forcing projections onto $V_\pm$; upper sign for A, lower sign for B.	Section 5.3
$\eta_t, \eta_\pm$	$\eta_t = 2\Psi^*\mathcal{Q}(\overline{\Phi}, p_{2\omega_0})$, $\eta_+ = 2\Psi^*\mathcal{Q}(\Phi, s_{2k_0})$, $\eta_- = 2\Psi^*\mathcal{Q}(\Phi, \overline{s_{2k_0}})$.	Section 6
η_x, a_+, γ	$\eta_x = \eta_+ = \eta_-$ for real S; $a_+ = a + \eta_x$, $\gamma = b + c$.	(57)
$z, \rho, f(\rho)$	$A = B = z$, $\rho = \lvert z\rvert^2$, $f(\rho) = \lvert a_+ + \gamma\rho\rvert^2 - \lvert\eta_t\rvert^2$.	(58)–(60)
ρ_*, ϕ_*, z_*	$f(\rho_*) = 0$, $\rho_* > 0$, $z_* = \sqrt{\rho_*}\mathrm{e}^{\mathrm{i}\phi_*}$; $\mathrm{e}^{-2\mathrm{i}\phi_*} = -(a_+ + \gamma\rho_*)/\eta_t$.	(61)
$v_\pm, J_\pm$	$v_\pm = \frac{1}{2}\mathrm{e}^{-\mathrm{i}\phi_*}(\delta A \pm \delta B)$; $J_\pm$ are their real 2×2 linearization matrices.	(65)–(67)

7 V1-inspired model and parameter choice

In this section we consider 4 types of populations: (E, PV, SST, VIP). For the connectivity matrix, rows index targets and columns index sources. Pfeffer et al. [1] motivate the retained inhibitory motif. Exact zeros are a reduction assumption. Billeh et al. [2] supply the cell-pair distance scales. Campagnola et al. [3] document the dependence of effective connectivity on cell class and layer.

Units, activation, and domain. Time is in ms, u, F, I are dimensionless, and

$$\ell_x = 100\,\mu\text{m}, \qquad L = 43, \qquad L_{\text{phys}} = 4.3\,\text{mm}, \qquad \tau = (10, 12, 35, 20)\,\text{ms}. \tag{71}$$

Retain the effective activation parameters

$$\begin{aligned} F_j(u) &= \ell(g_j(u-\theta_j)) - \ell(-g_j\theta_j), \qquad \ell(v) = (1+\mathrm{e}^{-v})^{-1}, \\ g &= (3.47682761, 7.34413013, 3.74650527, 6), \\ \theta &= (.28483799, .12068148, .40137356, .7). \end{aligned} \tag{72}$$

Literature-based widths and fixed gains. Billeh et al. [2], STAR Methods, p. e11, give $s_{E\to E} = 114$, $s_{E\to PV} = 92$, $s_{E\to SST} = 103$, $s_{PV\to E} = 95$, and $s_{SST\to E} = 85\,\mu$m, based on Levy and Reyes (2012), and $s_{PV\to PV} \simeq 120\,\mu$m from internal Allen Institute data. They use the PV–PV scale for other inhibitory pairs and the E–SST scales for E–Htr3a pairs. Replacing Htr3a by VIP is our additional reduction. Their distance rule is $\exp(-r^2/s_{ij}^2)$. Hence s_{ij} is the exponential distance scale and the standard deviation in our convention is $s_{ij}/\sqrt{2}$. With target rows and source columns,

$$s = \begin{pmatrix} 114 & 95 & 85 & - \\ 92 & 120 & 120 & - \\ 103 & - & - & 120 \\ 103 & - & 120 & - \end{pmatrix} \mu\text{m}, \quad \Sigma = \begin{pmatrix} .6498 & .45125 & .36125 & - \\ .4232 & .72 & .72 & - \\ .53045 & - & - & .72 \\ .53045 & - & .72 & - \end{pmatrix}, \quad \Sigma_{ij} = \frac{s_{ij}^2}{2\ell_x^2}. \tag{73}$$

Here Σ is the variance in model coordinates and $\Sigma^{\text{phys}} = \ell_x^2\Sigma$, with $\ell_x = 100\,\mu$m. A dash denotes an absent connection ($G_{ij} = 0$). For the effective integrated gains, we use

$$G = \begin{pmatrix} 6.22009 & -1.45455 & -6.48944 & 0 \\ 5.09798 & -.90615 & -8.04066 & 0 \\ 2.67174 & 0 & 0 & -16 \\ 1.68844 & 0 & -3.13584 & 0 \end{pmatrix}. \tag{74}$$

These magnitudes are fixed model coefficients. They are not measured connection probabilities.

Increase the scalar κ in $J = \kappa G$ from zero, compute its first modal crossing κ_c, and set

$$p = \kappa - \kappa_c, \qquad J(p) = (\kappa_c + p)G, \qquad J_0 = \kappa_c G, \quad J_1 = G. \tag{75}$$

The periodic-domain computation gives

$$\kappa_c = .7491489065, \qquad k_0 = 3, \qquad q_0 = \frac{6\pi}{43}, \qquad \omega_0 = .0829864799\,\text{rad/ms}. \tag{76}$$

Thus $f_0 = 13.2077085\,$Hz and the wavelength is $4.3/3$ mm.

Input populations. For visual/thalamic input, take $e = (1, 1, 0, 0)^{\mathsf{T}}$. Direct E and PV innervation is supported by Ji et al. [4] and used in Billeh et al.'s feedforward model. Equal normalized strengths are our assumption, not a measured ratio.

7.1 Computation of the amplitude coefficients

All quantities below are evaluated at onset $p = 0$, with $q_0 = 6\pi/43$ and ω_0 given in (76). With dimensionless amplitudes and $\tau = \epsilon^2 t$ measured in ms, $\alpha, a, b, c, c_\perp$ and all forcing coefficients η have units ms^{-1}. Matrices M, M_p and the multilinear maps Q_q, C_q likewise have units ms^{-1} on dimensionless arguments; $\mathcal{R}$ has units ms, and the response vectors h, $p_{2\omega_0}$, s_{2q_0} are dimensionless.

1. Activation derivatives and Fourier maps. Set $r_j = \ell(-g_j\theta_j)$. Direct differentiation gives

$$\begin{aligned} F_j'(0) &= g_j r_j(1-r_j), \\ F_j''(0) &= g_j^2 r_j(1-r_j)(1-2r_j), \\ F_j'''(0) &= g_j^3 r_j(1-r_j)(1-6r_j+6r_j^2). \end{aligned} \tag{77}$$

Form $D_F = \mathrm{diag}(F_j'(0))$ and $\boldsymbol{F}_r = (F_j^{(r)}(0))_j$, $r = 2, 3$. Substitute $\widehat{W}(q;0) = \kappa_c G \odot \mathrm{e}^{-\Sigma q^2/2}$ into (89) and (**??**) to obtain $M(q,0)$, Q_q, and C_q.

2. Critical eigenvectors and linear coefficient. Solve

$$M(q_0,0)\Phi = \mathrm{i}\omega_0\Phi, \qquad M(q_0,0)^*\Psi = -\mathrm{i}\omega_0\Psi, \qquad \|\Phi\|_2 = 1, \quad \Psi^*\Phi = 1. \tag{78}$$

Then

$$\alpha = \Psi^* D_\tau^{-1}(G \odot \mathrm{e}^{-\Sigma q_0^2/2}) D_F \Phi, \qquad a = \mu\alpha. \tag{79}$$

The unit-norm convention fixes the amplitude scale of b, c.

3. Quadratic responses. With $\mathcal{R}(\Omega, q) = (\mathrm{i}\Omega I_K - M(q,0))^{-1}$, solve

$$\begin{aligned} h_{20}^+ &= \mathcal{R}(2\omega_0, 2q_0) Q_{2q_0}(\Phi,\Phi), & h_{20}^0 &= 2\mathcal{R}(2\omega_0, 0) Q_0(\Phi,\Phi), \\ h_{11}^0 &= 2\mathcal{R}(0,0) Q_0(\Phi,\overline{\Phi}), & h_{11}^+ &= 2\mathcal{R}(0, 2q_0) Q_{2q_0}(\Phi,\overline{\Phi}). \end{aligned} \tag{80}$$

Each Q_q uses the output wavenumber of that interaction. Reflection symmetry gives the corresponding negative-wavevector responses.

4. Cubic coefficients. Project the resonant terms at output wavenumber q_0:

$$\begin{aligned} b &= \Psi^* \left[2Q_{q_0}(\Phi, h_{11}^0) + 2Q_{q_0}(\overline{\Phi}, h_{20}^+) + 3C_{q_0}(\Phi,\Phi,\overline{\Phi}) \right], \\ c &= \Psi^* \left[2Q_{q_0}(\Phi, h_{11}^0) + 2Q_{q_0}(\overline{\Phi}, h_{20}^0) + 2Q_{q_0}(\Phi, h_{11}^+) + 6C_{q_0}(\Phi,\Phi,\overline{\Phi}) \right]. \end{aligned} \tag{81}$$

The computed values, in ms^{-1}, are

$$\alpha = .0880577 + .0643807\mathrm{i}, \qquad b = -6.706479 + 9.069454\mathrm{i}, \qquad c = -6.961796 + 8.999612\mathrm{i}. \tag{82}$$

5. Forcing coefficients and input strength. For an input $I = \epsilon^2(P\mathrm{e}^{2\mathrm{i}\omega_0 t} + S\mathrm{e}^{2\mathrm{i}q_0 x} + \mathrm{c.c.})$ in (86), compute

$$p_{2\omega_0} = \mathcal{R}(2\omega_0, 0) D_\tau^{-1} P, \qquad s_{2q_0} = \mathcal{R}(0, 2q_0) D_\tau^{-1} S \tag{83}$$

$$\eta_t = 2\Psi^* Q_{q_0}(\overline{\Phi}, p_{2\omega_0}), \quad \eta_+ = 2\Psi^* Q_{q_0}(\Phi, s_{2q_0}), \quad \eta_- = 2\Psi^* Q_{q_0}(\Phi, \overline{s_{2q_0}}). \tag{84}$$

Here P, S are dimensionless input harmonics. The factors D_τ^{-1} convert them to rates in ms^{-1} in the u_t equation. For real S, $\eta_+ = \eta_- =: \eta_x$.

Let the default E/PV drive $e = (1, 1, 0, 0)^\mathsf{T}$. Take $\mu = -1$, $\epsilon = .02$, hence $p = -.0004$. The forcing coefficients are

$$\eta_t^{(1)} = .055509 - .029075\mathrm{i}, \quad \eta_x^{(1)} = .018902 + .046191\mathrm{i} \quad (\mathrm{ms}^{-1}). \tag{85}$$

8 Dispersion matrix and frequency variation

8.1 Fourier matrix for K populations

On $\mathbb{T}_L = \mathbb{R}/(L\mathbb{Z})$, consider the source-nonlinearity field

$$D_\tau u_t = -u + W(p) * F(u) + I(x,t), \qquad D_\tau = \mathrm{diag}(\tau_1, \dots, \tau_K), \quad \tau_i > 0, \tag{86}$$

where $F(u) = (F_j(u_j))_{j=1}^K$ is the activation function, $F(0) = 0$, and rows of W index targets, columns index sources. Use periodized, unit-mass Gaussian kernels of variances $\Sigma_{ij} > 0$:

$$W_{ij}(x;p) = J_{ij}(p) \sum_{\ell \in \mathbb{Z}} \frac{\mathrm{e}^{-(x+L\ell)^2/(2\Sigma_{ij})}}{\sqrt{2\pi\Sigma_{ij}}}, \qquad J(p) = J_0 + pJ_1. \tag{87}$$

Their convolution multipliers, with $q_h = 2\pi h/L$, are

$$\widehat{W}_{ij}(q;p) = J_{ij}(p)\mathrm{e}^{-\Sigma_{ij}q^2/2}, \qquad q = |q_h|, \quad h \in \mathbb{Z}. \tag{88}$$

Here $\widehat{W}$ is the integral Fourier transform of the kernel on the torus, without the factor L^{-1}.

Set $D_F = \mathrm{diag}(F_1'(0), \dots, F_K'(0))$. Linearizing at $u = 0$, $I = 0$ and substituting $u = v\mathrm{e}^{\nu t + \mathrm{i}q_h x}$ gives

$$\nu D_\tau v = -v + \widehat{W}(q;p) D_F v, \qquad \boxed{M(q,p) = D_\tau^{-1}\left[-I_K + \widehat{W}(q;p)D_F\right].} \tag{89}$$

Thus the dispersion relation is

$$\mathcal{P}(\nu, q, p) := \det(\nu I_K - M(q,p)) = 0. \tag{90}$$

The parameter p is measured from onset, as in the preceding sections. For $L = 2\pi$, $M_0(h) = M(|h|, 0)$ and $M_1(h) = \partial_p M(|h|, 0)$.

8.2 Eigenvalue derivatives

Fix p. Let $\nu(q)$ be a simple eigenvalue with right and left eigenvectors

$$Mv = \nu v, \qquad w^*M = \nu w^*, \qquad w^*v = 1. \tag{91}$$

Differentiation gives $M_q v + Mv_q = \nu_q v + \nu v_q$. Multiplying by w^* cancels the terms containing v_q, hence

$$\boxed{\nu_q = w^* M_q v, \qquad \nu_p = w^* M_p v,} \tag{92}$$

where, with entrywise exponentiation,

$$\begin{aligned} M_q &= -qD_\tau^{-1}\left[(\Sigma \odot \widehat{W}(q;p))D_F\right], \\ M_p &= D_\tau^{-1}\left[(J_1 \odot \mathrm{e}^{-\Sigma q^2/2})D_F\right]. \end{aligned} \tag{93}$$

The second identity assumes that F and D_τ do not depend on p. At the critical point, $v = \Phi$, $w = \Psi$, and $\nu_p = \alpha$.

Write $\nu = \sigma + \mathrm{i}\omega$ with $\omega > 0$. Then

$$\sigma_q = \mathrm{Re}(w^* M_q v), \qquad \boxed{\omega_q = \mathrm{Im}(w^* M_q v).} \tag{94}$$

Equivalently, since $\mathcal{P}(\nu(q), q, p) = 0$ holds identically along the branch, implicit differentiation of (90) gives

$$\nu_q = -\frac{\mathcal{P}_q}{\mathcal{P}_\nu}, \qquad \boxed{\omega_q = \frac{\operatorname{Im}(\mathcal{P}_\nu \overline{\mathcal{P}_q})}{|\mathcal{P}_\nu|^2},} \tag{95}$$

with all derivatives of $\mathcal{P}$ evaluated at $(\nu(q), q, p)$. Because ν is a simple root, $\mathcal{P}_\nu \neq 0$, so the sign of ω_q is the sign of the numerator $\operatorname{Im}(\mathcal{P}_\nu \overline{\mathcal{P}_q})$.

8.3 The rightmost eigenvalue

The formulas above follow a single eigenvalue branch. The frequency that matters for the instability is that of the eigenvalue with the largest real part. We therefore define the spectral abscissa $\sigma_{\max}$ and the frequency $\omega_{\mathrm{r}} \geq 0$ of the rightmost eigenvalue by

$$\sigma_{\max}(q) = \max_{\nu \in \operatorname{spec} M(q,p)} \operatorname{Re} \nu, \qquad \omega_{\mathrm{r}}(q) = \max \left\{ \operatorname{Im} \nu : \operatorname{Re} \nu = \sigma_{\max}(q) \right\}. \tag{96}$$

Since M is real, complex eigenvalues come in conjugate pairs. If a pair is rightmost, ω_{r} is its positive imaginary part. If a unique real eigenvalue is rightmost, $\omega_{\mathrm{r}} = 0$.

V1 parameters. For the parameters of Section 7, at $p = 0$ and $q_0 = 6\pi/43$,

$$\omega_q(q_0) = .0471593 > 0, \qquad \frac{d\omega}{dq_{\mathrm{phys}}} = .00471593 \,\mathrm{mm/ms}, \tag{97}$$

where q is the model-coordinate wavenumber and ω is in rad/ms. The rightmost frequencies of modes $k = 0, 1, 2, 3, 4$ are 11.0865, 11.3759, 12.1588, 13.2077, and 14.2833 Hz, so the frequency increases across these modes; this does not imply global monotonicity. Changing L changes which q values are sampled, but not the derivative at fixed q and κ.

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